Chapter 1: Cell Structure

Key Outcomes

By the end of this chapter, you should be able to:

1.1. Understand that cells are the basic unit of life

1.2. Use the units of measurements relevant to Light microscopy and understand basic conversions

1.3. Calculate magnifications and actual sizes of specimen using drawings or micrographs

1.4. Understand the use of electron microscope to study cells

1.5. Recognise and identify common structures & organelles found in eukaryotic cells and their functions

1.6. Compare structural features of animal cells and plant cells

1.7. Outline briefly the role of ATP in cells

1.8. Describe the structure of bacteria and prokaryotes vs eukaryotes

1.9. Describe the structure of viruses

1.1. Introduction

- Cells are the basic unit of life, they every living thing is made up of a wide collection of cells, and every cell comes from pre-existing cells by the process of cell division.
- A cell is like a bag with a lot of chemical reactions going on inside it, that is therefore separated from the environment outside the cell, as this cell is surrounded by a thin membrane.
- This thin membrane is an essential feature of the cell as it acts like a barrier, control the exchange between outside and inside the cell, letting in some essential substances to the cell and blocking out what the cell does not need, hence why it's described as partially-permeable.
- All types of cells have genetic material (DNA) the more complex cells have it inside of a nucleus, and the simpler cells have it free in the cytoplasm. All cells also have organelles, which are the functioning structures inside the cell (e.g. mitochondria)

1.2. The Light Microscope (LM):



- To study cell biology, scientists cannot see the cells with their naked eye, so they must use something with greater magnification and higher resolution, like Light or Electron microscopes
- The light microscope uses light as a source of radiation to illuminate the cells, which are much smaller than regular measurement units we are used to, so we use:

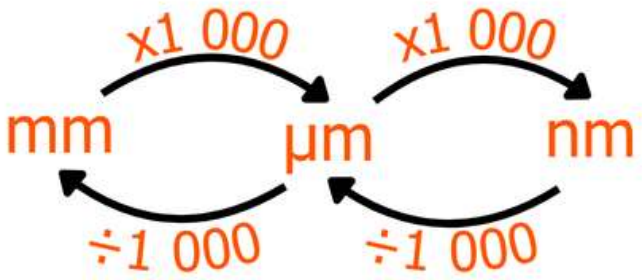
Fraction of mm	Units	Symbols
1/1 000 mm	micrometer	µm
1/1 000 000 mm	nanometer	nm

1. Exercise:

0.02mm = ...µm

350nm = ...µm

0.0045mm = ...nm



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The smallest thing that a human eye can see is about 50-100μm, any two objects with a distance separating them smaller than that, both objects will be seen as one, indistinguishable, while cells in the human body vary from 5 to 40μm.

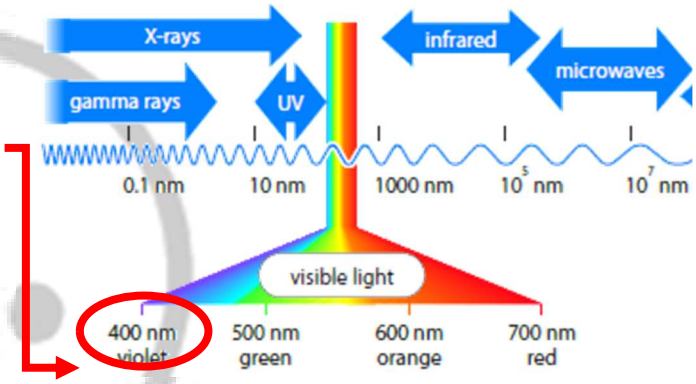
While the smallest thing a Light Microscope can see is about 200nm (0.2μm), this is known as the Resolution of the LM

Resolution: it's the minimum distance between two particles that allows them to be distinguished as two separate objects

- In a light microscope, Light is used as source of radiation

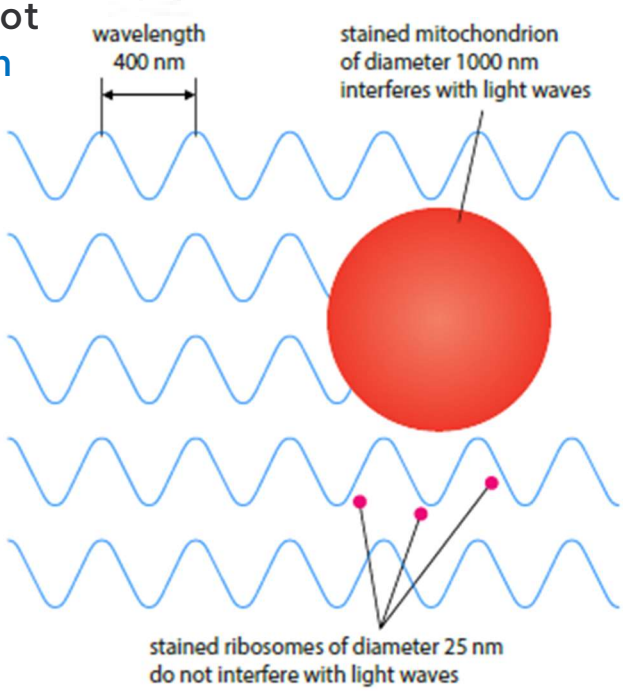
For any object to be visible to the human eye, light waves must be able to interfere with it, so the object must be:

1. Larger than ½ the lowest wavelength of visible light (>½ 400nm = >200nm)
2. Not be transparent and let light waves pass through it



Therefore, the 1000nm mitochondria can be seen under the LM, as it can interfere with the light waves (1000nm > 200nm)

Whilst the much smaller 25nm ribosome cannot be seen, as they are too small to interfere with the light waves (25nm < 200nm)



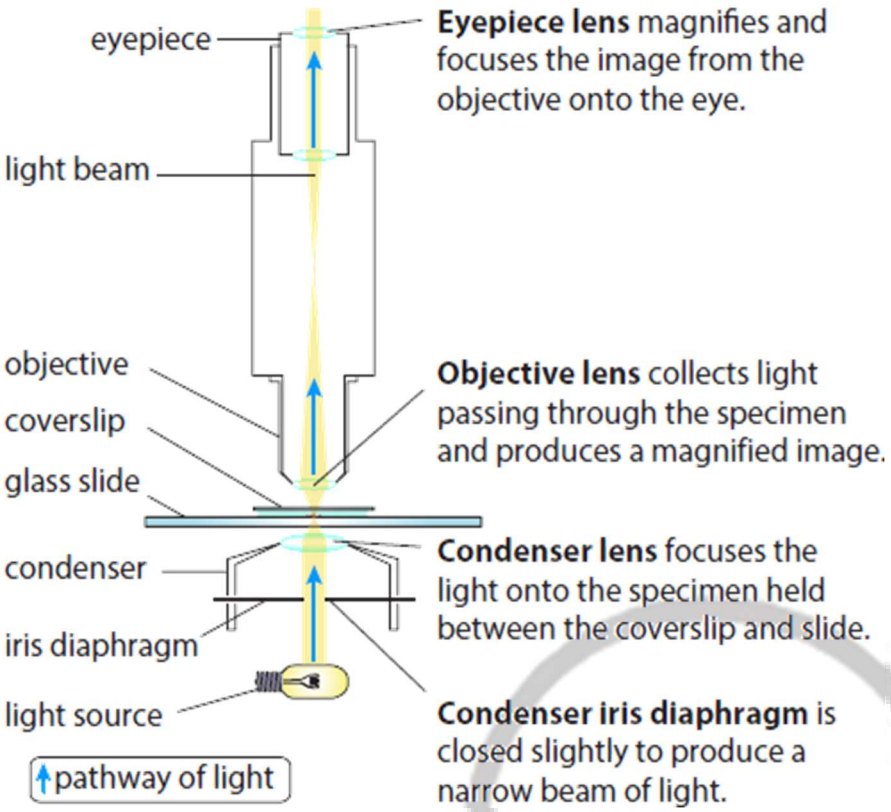
2. Explain why ribosomes cannot be seen using a LM

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General structure of LM



NOTE

Most slides under LM have to be stained, as most structures are naturally transparent

NOTE

In resolution, lower number is a higher resolution,

Higher resolution = more detail

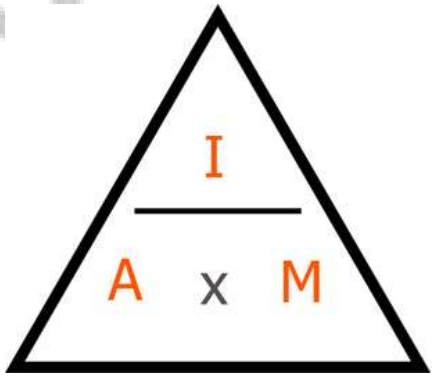
1.3. Measurement of structures using the Light microscope:

Magnification: is the number of times larger an image of an object is than its actual real size

- Magnification can be changed on the LM using the different objective lenses

To measure the structures, we use the I AM rule:

- Image size = Actual x magnification
- Actual size = Image / magnification
- Magnification = Image / Actual



Measuring using eyepiece graticule:

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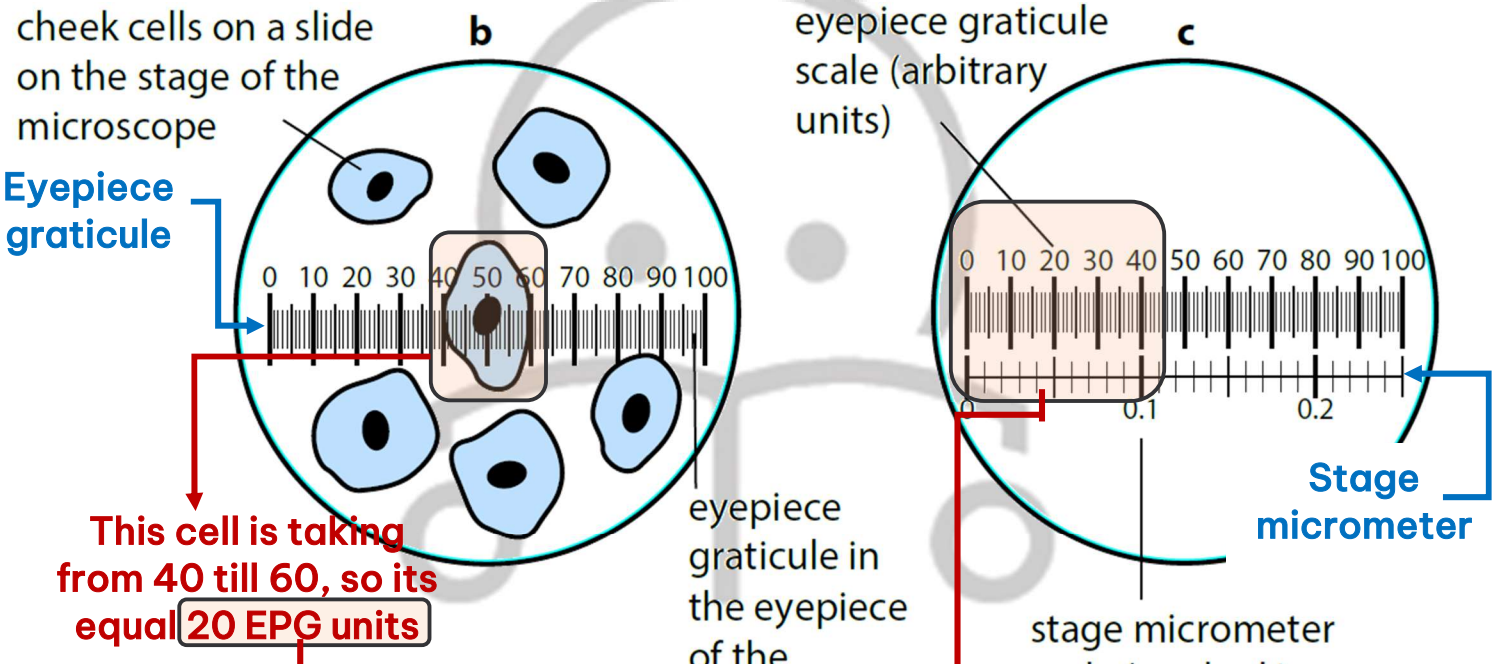
Eyepiece graticule is a small scale usually with 100 divisions that is placed on the eyepiece of the microscope for measuring structures

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* However, 1 eyepiece graticule unit (EPG) does not equal a certain length by default, **unlike a normal ruler**/scale, as 1 EPG unit equals different lengths when different objective lenses are used, so it must first be calibrated *

Calibration of an eyepiece graticule:

- A tiny transparent ruler called a stage micrometer is placed on the microscope slide
- The stage micrometer usually has very accurate tiny subdivisions of 0.01mm and 0.1mm
- The eyepiece graticule scale and the stage micrometer scale are then aligned to be on top of one another then the length of 1 EPG unit is found out



In this case:

x1000

Step 1: 40 EPG = 0.1mm → 40 EPG = 100µm

Step 2: 1 EPG = ??

Step 3: ?? = 100µm/40EPG → 1 EPG = 2.5µm

Step 4: Cell = 20 EPG x 2.5µm → Cell = 50µm

1.4. The Electron microscope (EM):

So, we now know how to see structures like the mitochondria, larger than 200nm, what about structures smaller than 200nm?

We also know that these structures need a different source of radiation, with lower wavelengths, using light is not suitable

Electrons shooting from very hot metals have very high energy and much lower wavelengths. Using electrons as radiation allows resolution of 0.2 nm, making them much more suitable

So, electron microscopes are the way to go, they show smaller structures in higher details, but give a non-living, black & white image for the specimen

Scanning Electron Microscope (SEM)

- Reflected electron beam is observed
- 3D external surfaces of structures can be seen
- Lower resolution (3-20nm)

Transmission Electron Microscope (TEM)

- Electron beam passing through specimen is observed
- 2D thin sections internal structures can be seen
- Higher resolution (0.2nm)

Light Microscope



- ✓ Specimen can be alive
- ✓ Shows colors
- ✓ Cheap & small
- ✓ Can be used by students
- - Lower resolution (200nm)
- - Lower magnification (x1500)

Electron microscope



- - Specimen must be dead
- - Black & white images
- - Expensive & large
- - Needs professionals
- ✓ Higher resolution (0.2nm)
- ✓ Higher magnification (x500 000)

Advantage

Disadvantage

Electron microscopes are INCREDIBLY LARGE

3. Exercise:

Fig. 1.1 is a transmission electron micrograph of a part of an animal cell.



magnification ×16 000

(a) Calculate the actual width of the organelle labelled **A**, as shown by line **X–Y**.

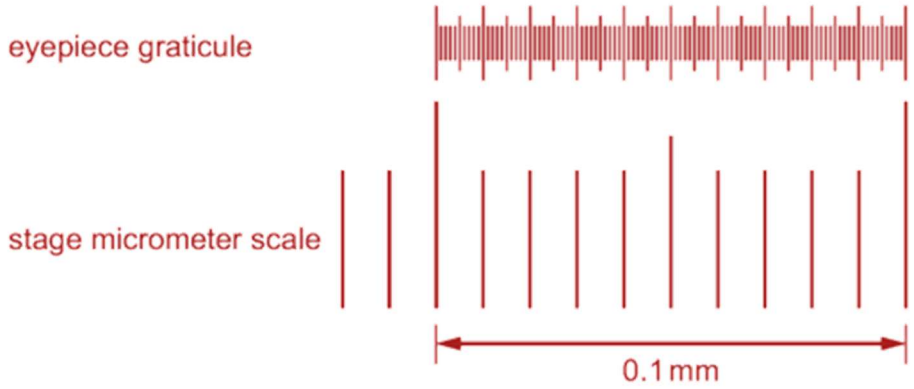
State the formula that you will use and show your working.

Give your answer in μm and to one decimal place.

formula

..... μm [3]

4. The diagram shows an eyepiece graticule and part of a stage micrometer scale as seen using ×100 magnification.



Which is the correct method for calculating the value of one eyepiece graticule unit in micrometres (μm)?

A divide 100 by 0.1 then multiply by 1000

B divide 100 by 0.1 then multiply by 1000 divided by 100

C multiply 0.1 by 1000 then divide by 100

D multiply 0.1 by 1000 then divide by 100 then divide again by 100

5. An eyepiece graticule can be calibrated using a stage micrometer.

What is the correct reason why an eyepiece graticule is calibrated?

A An eyepiece graticule can be used to make measurements.

B An eyepiece graticule is magnified by the objective lens.

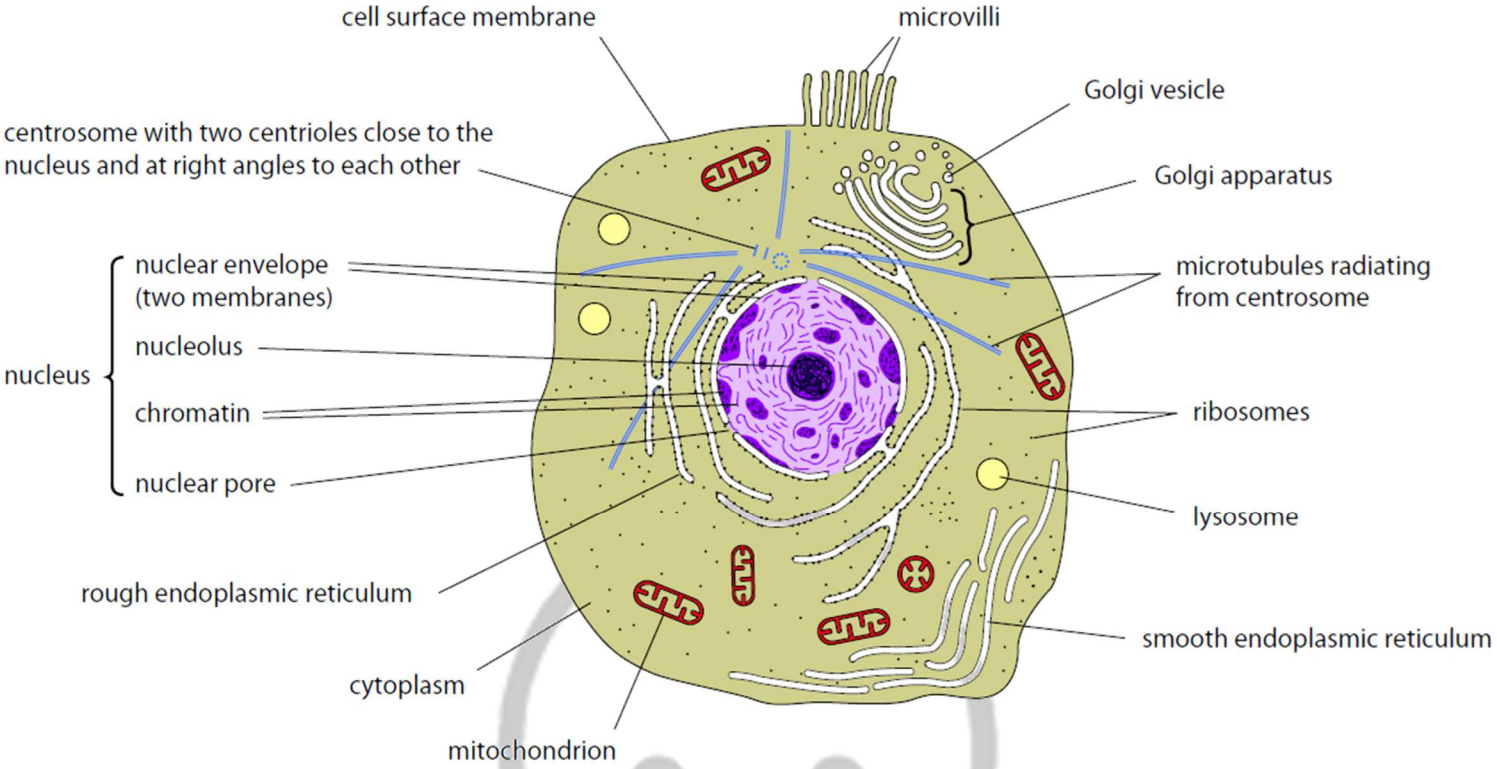
C An eyepiece graticule magnifies the specimen.

D An eyepiece graticule makes comparisons.

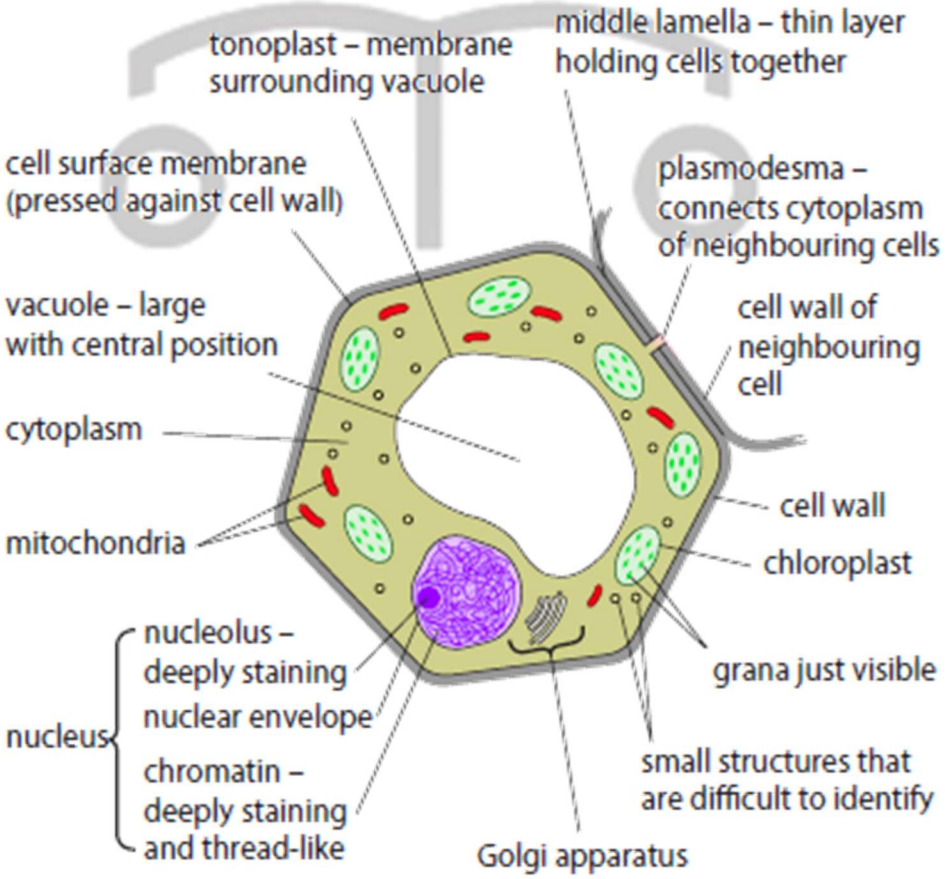
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1.5. Eukaryotic Cell Structure

This diagram shows what an animal cell looks like under an EM:



And this is a plant cell under a high-quality LM:



Now, there are many things that we know about these 2 from Biology Olevel, lets dive into greater detail now

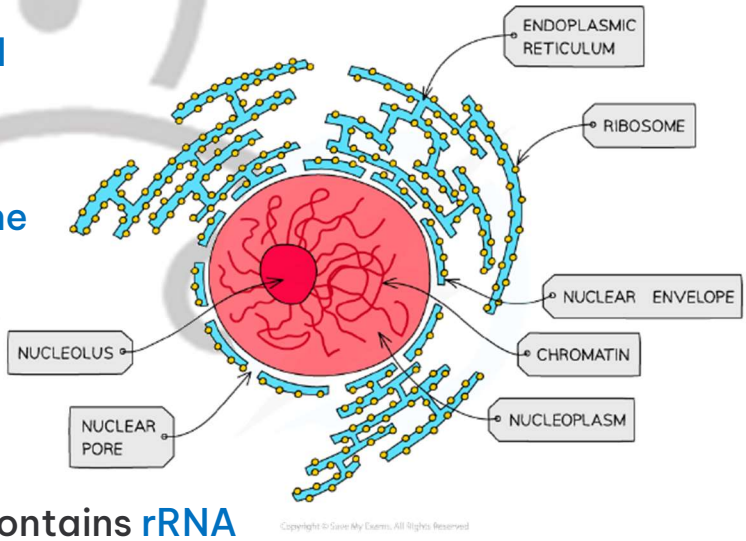
1.6. Structures found in animal and plant cells:

1. Cell surface membrane (7nm):

- All cells are surrounded by a very thin cell surface membrane
- Controls substance exchange between cell and environment
- Barrier between internal and external environment for cells
- Anchors the Cytoskeleton
- Made of phospholipids, cholesterol and proteins (Covered in Ch4)

2. Nucleus (relatively large)

- All eukaryotic cells have a nucleus
- DNA inside the nucleus is found in the form of Chromatin, which are threads of DNA coiled around Histone proteins that collect to form Chromosomes (Covered in ch5)
- DNA has the genes that code for all proteins to be synthesised
- Surrounded by a nuclear envelope, which is a double layered membrane that has many small nuclear pores that allow exchange of substances between cytoplasm & nucleus



- Nucleolus is another structure in the nucleus that is responsible for production of ribosomes, and contains rRNA
- mRNA produced from the DNA in chromosomes travels out the nuclear pores into the cytoplasm to the ribosomes on RER, which is why RER are associated with the outer membrane of the nuclear envelope (see Ch6)

An envelope → double layered membrane (inner membrane and outer membrane, with intermembrane space), e.g. nuclear envelope

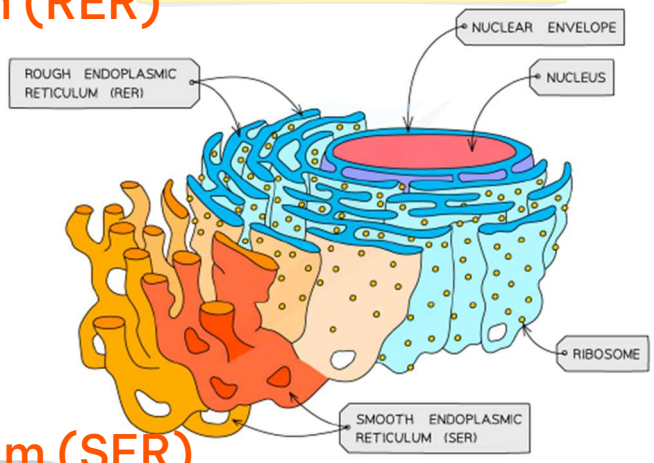
A membrane → Single layer membrane, e.g. any other membranous organelle

3. Rough & smooth endoplasmic reticulum

- Made of a network of membranes forming flattened hollow sacs called cisternae

Can only be seen under EM

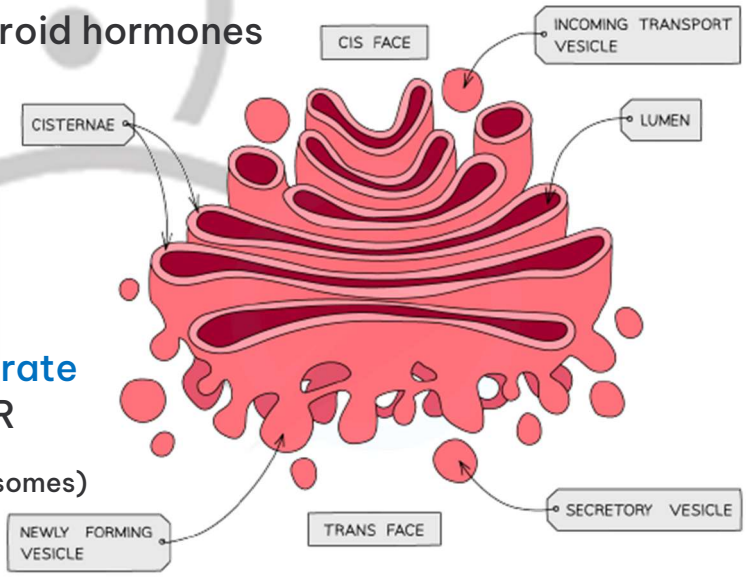
- **Rough endoplasmic reticulum (RER)**
  - Continuous with the nuclear envelope
  - Covered with ribosomes
  - Site for protein synthesis
  - Packs proteins in vesicles to send them to Golgi body



- **Smooth endoplasmic reticulum (SER)**
  - Does not have ribosomes on surface, not continuous to nuclear envelope
  - Is more tubular and irregular
  - Production of lipids and steroid hormones

2. Golgi Body

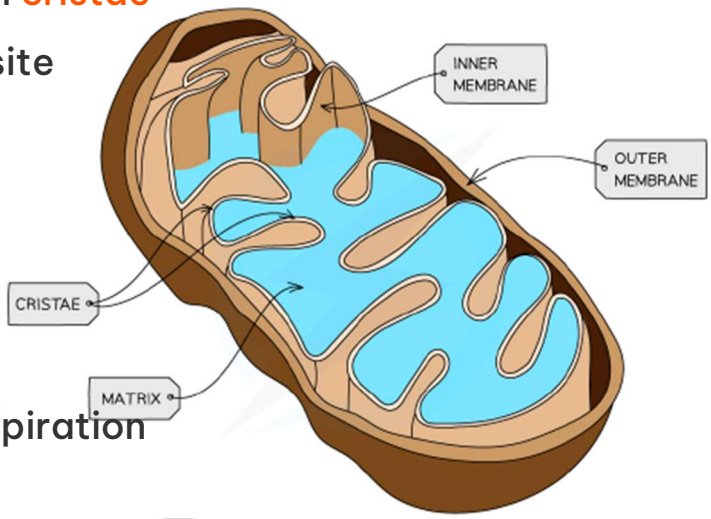
- Stacked flattened sacs of membrane, also have cisternae
  - HOWEVER, its sacs are separate not interconnected, unlike ER
  - Its smooth, unlike RER (no ribosomes)
  - Its close to nucleus
  - Has 2 faces (Cis & trans)



- Site for folding of proteins & Post-transational modifications (e.g., glycosylation (see 4.2))
- Recieves transport vesicles from RER, and Packages proteins into secretory vesicles for exocytosis
- Site for formation of lysosomes
- Involved in synthesis of new plant cell walls

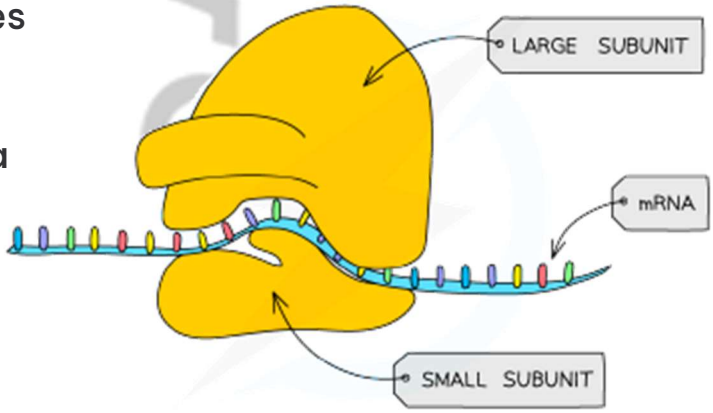
3. Mitochondria

- Relatively large in diameter, similar size to prokaryote cells (~0.5-1μm)
- Surrounded by a double-membrane (inner, outer)
- Inner membrane is folded to form cristae
- Inner membrane and matrix are site for Aerobic respiration and formation of ATP
- Inside the matrix there is circular DNA (mitochondrial DNA) and 70s ribosomes (explained later)
  - For production of aerobic respiration enzymes
- Mitochondria can divide on their own, synthesise proteins on their own, so they can function independently from the cell



4. Ribosomes (25nm)

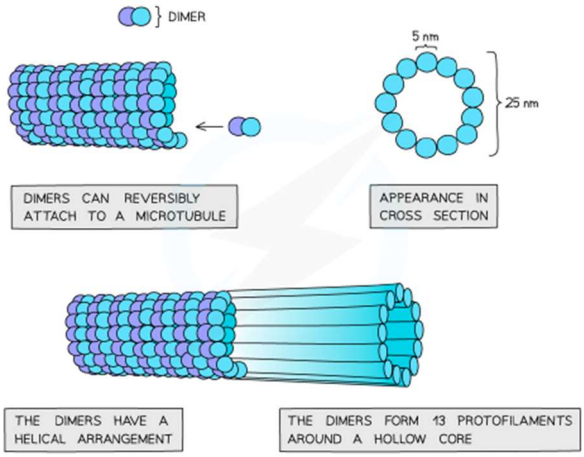
- Ribosomes are non-membranous organelles found in cytoplasm of all cells, or as part of RER in eukaryotes
- Made of rRNA and proteins
- Composed of a larger subunit and a smaller subunit
  - Small 70S ribosomes are found in prokaryotes, mitochondria and chloroplasts
  - Larger 80S ribosomes are found in eukaryotic cells
- They are the site of translation during protein synthesis (Covered in Ch6)



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5. Microtubules (25nm)

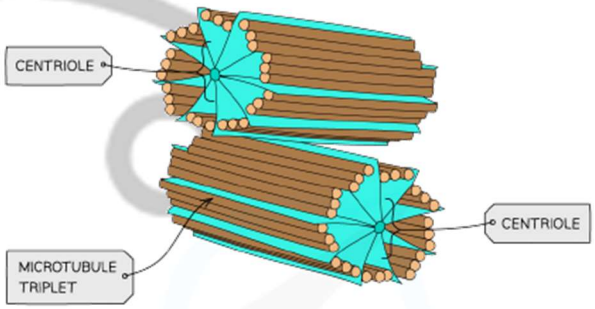
- Microtubules are hollow non-membranous tubules made of tubulin protein
  - α and β tubulin proteins combine to form dimers, which are then joined into protofilaments
  - 13 protofilaments in a cylinder make a complete microtubule



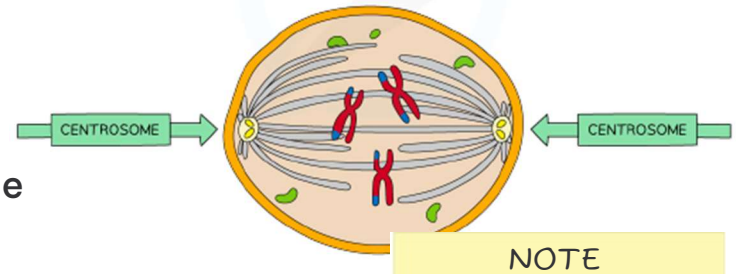
- Microtubules & Actin microfilaments make up the cytoskeleton of the cell
- Cytoskeleton provides support and guide movement of vesicles inside the cell, anchored by the cell membrane
- Microtubules are made in Microtubule organizing centres (MTOCs) in plant cells, and in centrosome in animal cells
- They form centrioles and spindle fibers that are essential for cell division (Covered in Ch5)

Structures only found in animal cells:

1. Centrioles & Centrosomes



- Hollow fibres made of nine triplets of microtubules
- Two centrioles found close to the nucleus at right angles of each other in a region called the centrosome
- The centrioles & centrosome migrates to the poles of the cell during cell division to organize the spindle fibres (Covered in Ch5)

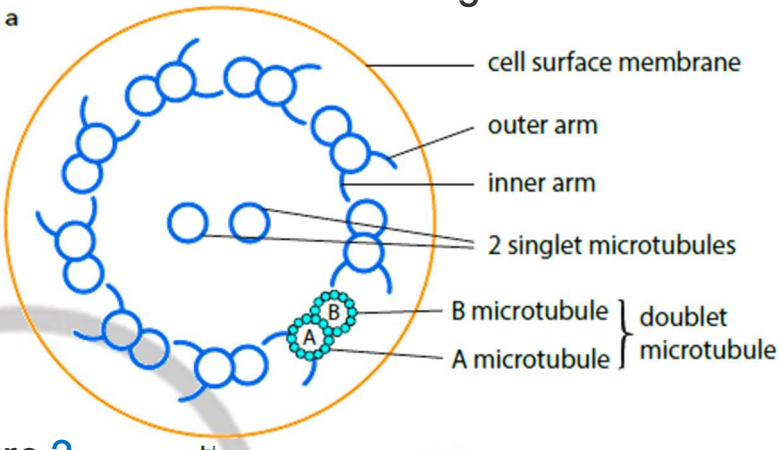


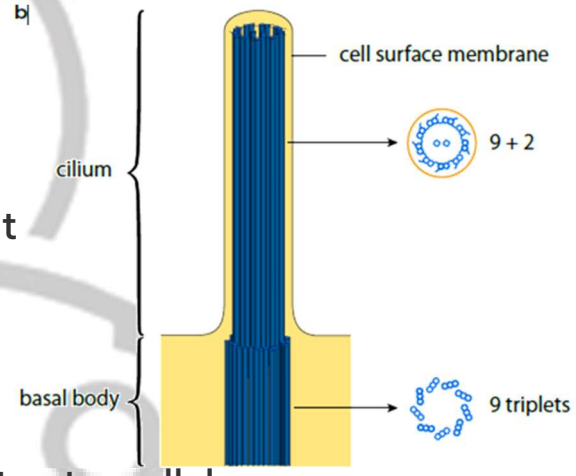
NOTE
NO CENTRIOLES IN PLANT CELLS

- The centrosome region is considered a microtubule organizing centre (MTOC), produces microtubules for cytoskeleton during cell rest, and produces microtubules for spindle fibres during cell division

2. Cilia

- Cilia are hair-like structures that extend from many types of eukaryotic cells, that are made of microtubules and over 600 polypeptides, in a very complex arrangement
- Their base is made of a centriole arrangement (9 triplets)
- Their shaft is made of 9 microtubule doublets with 2 singlets in the center (9+2 arrangement)
  - Each doublet is made of one complete ring with 13 protofilaments called microtubule A, and one incomplete ring with only 10 protofilaments called microtubule B
  - Attached to microtubule A are 2 arms made of protein dynein, one inner arm and one outer arm





- This complex structure allows them to beat very accurately to sweep mucus out of the respiratory tract
- The entire cilium shaft is surrounded by an extension of the cell membrane, so cilia are considered an intracellular structure, not extracellular

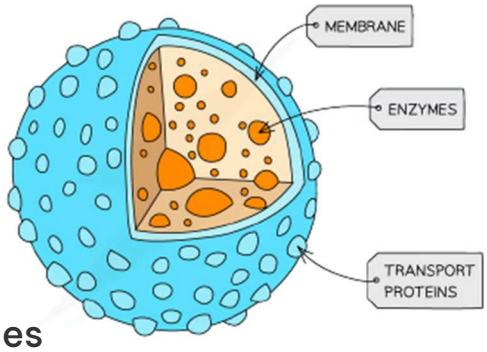
3. Microvilli

- Extracellular Hair like structures made of Actin microfilaments found in some types of eukaryotic cells
- They increase surface area of cells for more absorption/diffusion

NOTE
Cilia and flagella have the same structure

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4. Lysosomes (0.1-0.5μm)

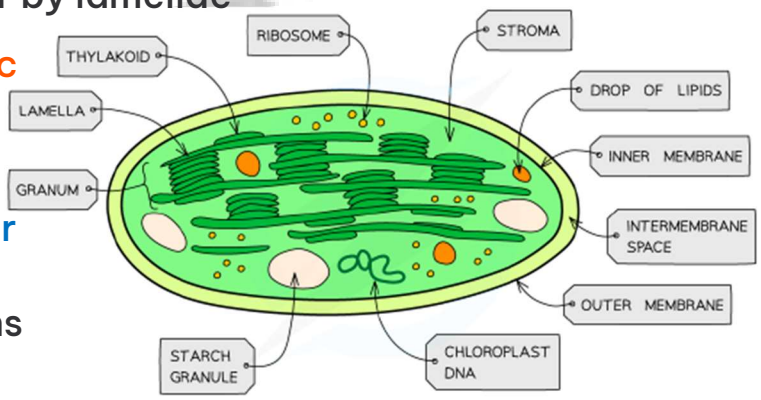


- Specialized vesicles surrounded by a single membrane
- Contain hydrolytic (digestive) enzymes to break down molecules like:
  - Waste materials such as dead/old organelles
  - Engulfed pathogens during phagocytosis/endocytosis
    - By fusing with the phagocytic vacuole formed and release the enzymes to digest the pathogens
  - Cell debris during apoptosis (programmed cell death)
- Contain 60+ types of enzymes like (proteases, lysozymes, lipases…)
- In plant cells, large central vacuole usually acts as lysosome

Structures only found in plant cells:

1. Chloroplasts

- Structure larger than mitochondria, surrounded by double membrane
- Contain membranous compartments called thylakoids
  - ➢ Thylakoids stack together to form grana
  - ➢ Many grana are joined together by lamellae
- Thylakoids contain photosynthetic pigments that absorb light energy for photosynthesis
- Chloroplasts contain small circular DNA and 70S ribosomes to synthesise photosynthetic proteins

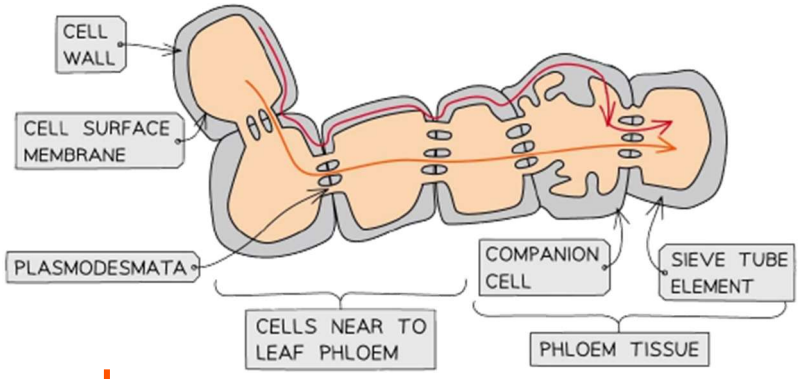


2. Cell wall

- Cell walls are outside the cell surface membrane and offer structural support, protect highly turgid cells from bursting
- Made by cellulose in plant cells, and by chitin in fungi
- They are freely permeable, do not play a role in control of substance exchange with cell membrane

3. Plasmodesmata

- Are bridges in cytoplasm between neighbouring plant cells, allowing the transfer of substances (Covered in Ch7)



4. Large central vacuole

- Allows plant cells to store cell sap and provide more structural support to the cells
  - ➢ Sometimes vacuoles are found in animal cells (e.g. phagocytic vacuole) but they are small and temporary
- Surrounded by a tonoplast which is a partially permeable membrane
- Have lysosomal activity, contain hydrolytic enzymes to break down pathogens or dead organelles
- Responsible for the majority of the increase in size of plant cells during growth

In Olevel, we have learned that mitochondria produce energy by aerobic respiration in the form of ATP

1.7. Now, what is ATP?

ATP (adenosine tri-phosphate) is the molecule that is the universal energy for currency in all living cells,

- Energy released from energy-releasing reactions is used to make ATP
- When energy requiring-reaction is occurring, ATP is broken down to ADP (Adenosine diphosphate) to release energy

$$ATP \rightleftharpoons ADP + Pi + Energy$$

- Energy released is needed in many cellular processes like:
  - Active transport
  - Muscle contraction
  - Protein synthesis
  - Endocytosis/exocytosis

6. Exercise: Fill this table out using ticks (✓) and crosses (✗)

Structures	Animal cell	Plant cell
Mitochondria		
Cell wall		
70S ribosomes		
Microtubules		
Endoplasmic reticulum		
Plasmodesmata		
Centrioles		

Domains are the highest level of organism classification; Eukaryotic or Prokaryotic

- Eukaryotes make up the 4 kingdoms we know (Plantea, Animalia, Fungi and Protoctista)
- Prokaryotes make up all types of bacteria and algae

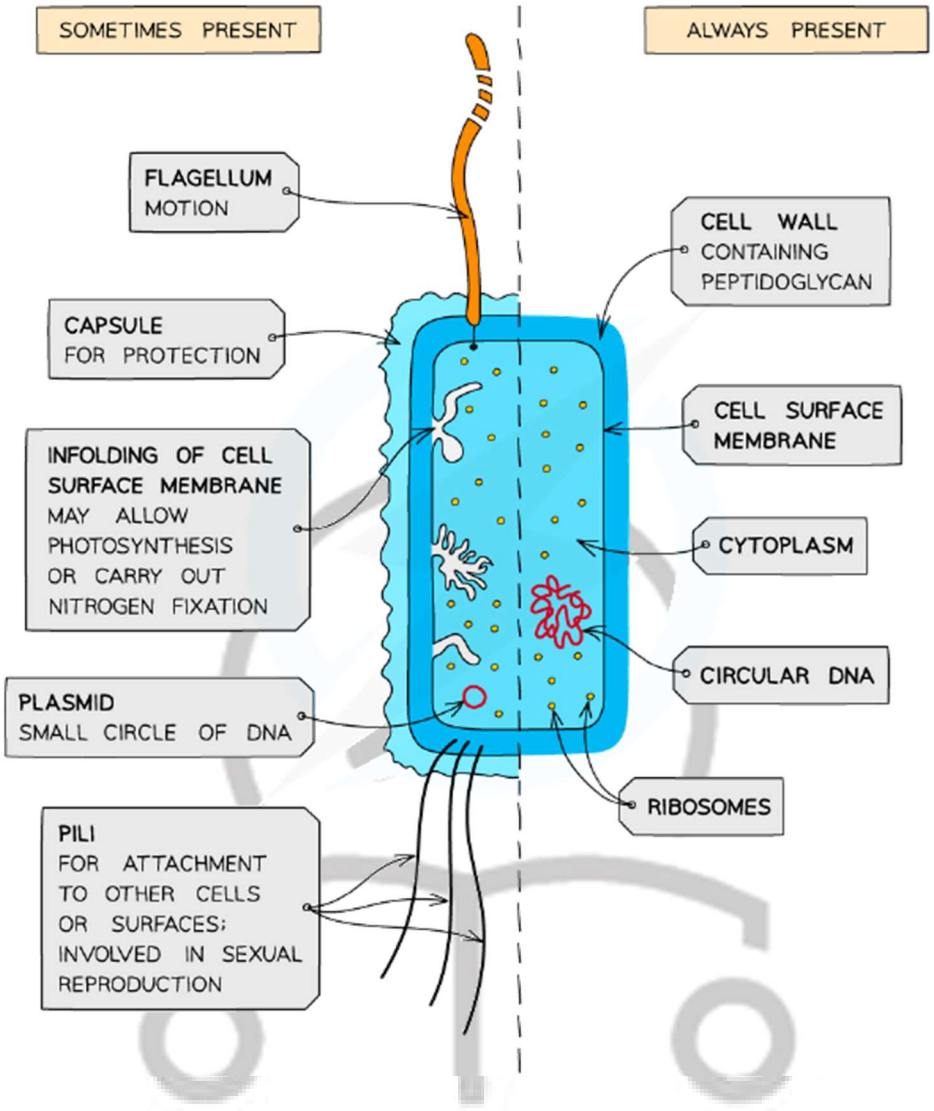
1.8. Prokaryotic cell structure

Always present:

- Circular DNA
- No membranous organelles
  - ONLY 70s Ribosomes
- Only 1-5 μm in diameter, similar to mitochondria
- Peptidoglycan cell wall
- Always unicellular
  - Eukaryotes can be unicellular (fungi) or multicellular (animals)
- Prokaryotes divide by binary fission, while eukaryotes by mitosis or meiosis
  - No spindle fibres involved for prokaryotes

Sometimes bacterial cells have:

- Flagella for movement
- Plasmids (small circular DNA)
- Capsules for protection



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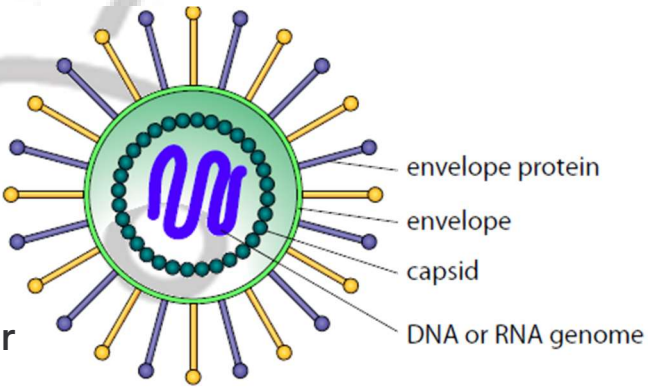
	Prokaryotes	Eukaryotes	
		Animal Cells	Plant Cells
Cell surface membrane	Always present	Always present	Always present
Cell wall	Always present; peptidoglycan	Never present	Always present; cellulose
Nucleus	Never present	Always present	Always present
Chromosomes	Never present	Always present	Always present
Chloroplasts	Never present	Never present	Sometimes present
RER & SER	Never present	Usually present	Usually present
Ribosomes	70S ribosomes	80S in cytoplasm, 70S in mitochondria	80S in cytoplasm, 70S in mitochondria and chloroplasts
Centrioles	Never present	Usually present	Never present

1.9. Viruses

Viruses are tiny non-living particles (20-300nm) much smaller than bacteria

All they consist of is:

- DNA or RNA genetic material
- Protective protein coat (Capsid)
- Capsid is made of many small protein subunits called 'Capsomeres'
    - Some viruses also contain an outer envelope made of phospholipids
- They are Acellular, non-living structures



Viruses depend on infecting and taking over other living cells, using the host cell's ribosomes and golgi body to make their own proteins from their DNA or RNA.

Textbook Style Questions

1. Which one of the following cell structures can be seen with a light microscope?

A mitochondrion

B ribosome

C rough ER

D smooth ER

2. What property of electrons allows high resolution to be achieved by electron microscopes?

A Electrons are negatively charged.

B Electrons can be focused using electromagnets.

C Electrons have a very short wavelength.

D Electrons travel at the speed of light.

3. Which one of the following structures is found in animal cells but not in plant cells?

A cell surface membrane

B centriole

C chloroplast

D Golgi apparatus

4. List ten structures you could find in an electron micrograph of an animal cell which would be absent from the cell of a bacterium.

5. Distinguish between the following pairs of terms:

a magnification and resolution

b light microscope and electron microscope

c nucleus and nucleolus

d chromatin and chromosome

e membrane and envelope

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f smooth ER and rough ER

g prokaryote and eukaryote

h cell wall and cell surface membrane

i capsid and cell wall

j capsid and capsomere

6 List:

a three organelles each lacking a boundary membrane

b three organelles each surrounded by a single membrane

c three organelles each surrounded by two membranes (an envelope).

7. Identify each cell structure or organelle from its description below.

a manufactures lysosomes

b manufactures ribosomes

c site of protein synthesis

d can bud off vesicles which form the Golgi apparatus

e can transport newly synthesised protein round the cell

f manufactures ATP in animal and plant cells

g controls the activity of the cell because it contains the DNA

h carries out photosynthesis

i can act as a starting point for the growth of spindle microtubules during cell division

j contains chromatin

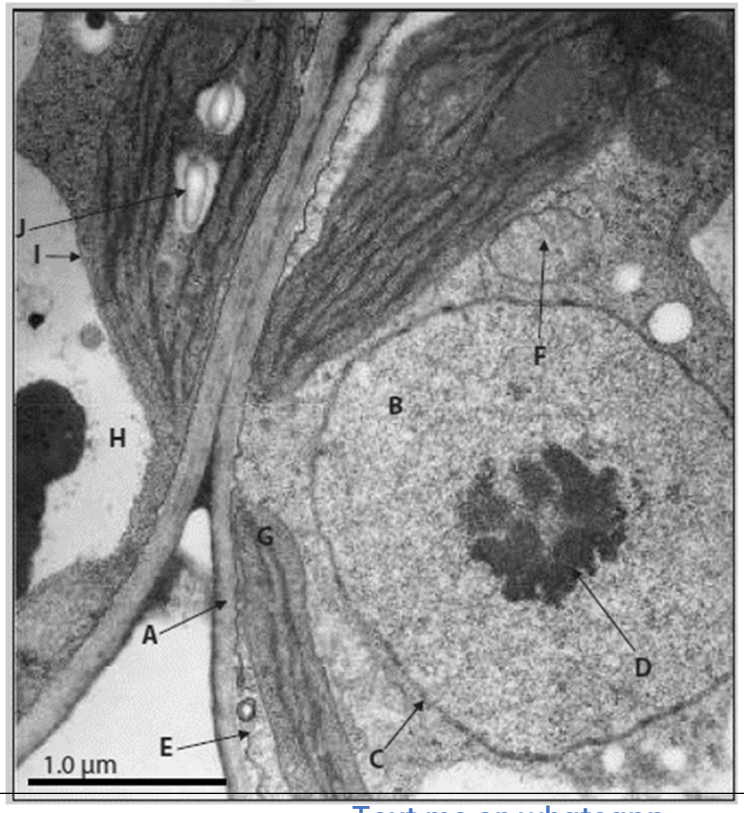
k partially permeable barrier only about 7 nm thick

l organelle about 25 nm in diameter

m organelle with a '9 + 2' arrangement of microtubules

8. The transmission electron micrograph shows parts of two palisade cells from a leaf

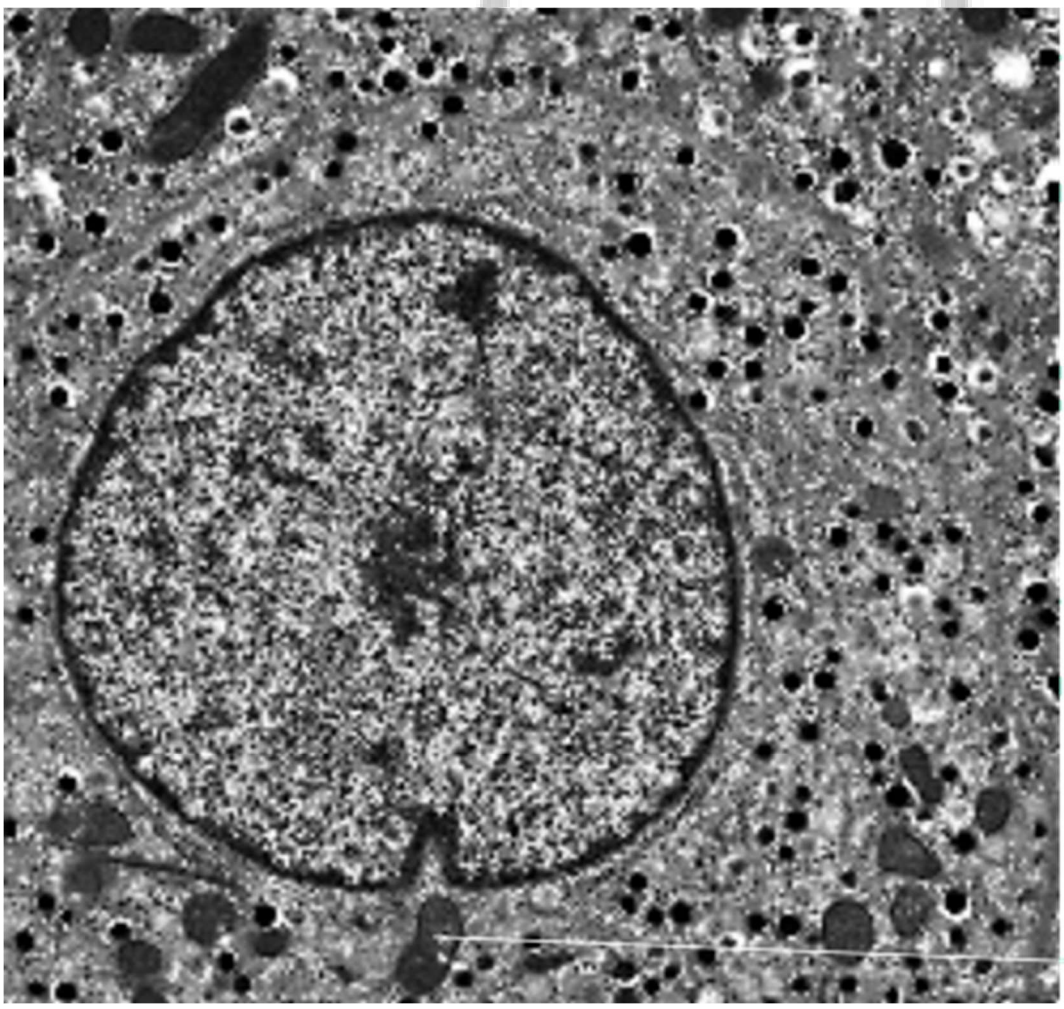
Identify the labelled structures A–J and write a brief statement about their functions.



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Label	Name of structure	Function	Label	Name of structure	Function
A			F		
B			G		
C			H		
D			I		
E			J		

9. The electron micrograph shows part of a secretory cell from the pancreas. You are not expected to have seen a micrograph of this type of cell before. The cell contains many secretory vesicles. These are Golgi vesicles. They appear as small, roughly circular structures with black circular contents. The magnification is ×8000.



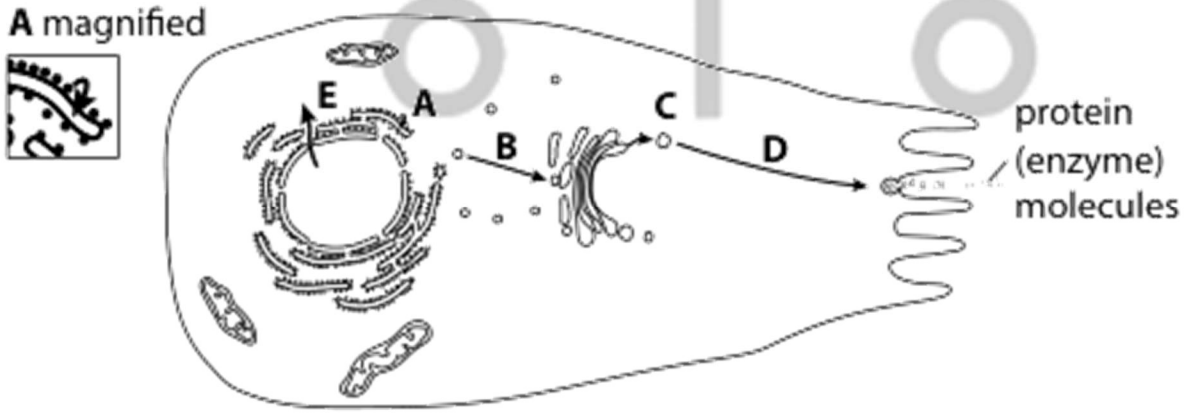
a Calculate the actual sizes of the structures listed in the table. Use a ruler with mm divisions to help you. Show your measurements and calculations. When you have your answers, complete the table with the required information. Give your answers in micrometres.

Structure	Observed diameter (measured with ruler)	Actual size
maximum diameter of a Golgi vesicle		
maximum diameter of nucleus		
maximum length of the labelled mitochondrion		

c The mitochondria in pancreatic cells are mostly sausage-shaped in three dimensions. Suggest why some of the mitochondria in the electron micrograph here appear roughly circular.

d The figure is a diagram based on an electron micrograph of a secretory cell from the pancreas. This type of cell is specialised for secreting (exporting) proteins. Some of the proteins are digestive enzymes of the pancreatic juice. The cell is very active, requiring a lot of energy. The arrows A, B, C and D show the route taken by the protein molecules.

Note that arrow A is shown magnified in a separate diagram.



i Describe briefly what is happening at each of the stages A, B, C and D.

ii Arrow E shows the path of a molecule or structure leaving the nucleus through the nuclear envelope. Name one molecule or structure which leaves the nucleus by route E.

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iii The molecule or structure you named in ii passes through the nuclear envelope. Name the structure in the nuclear envelope through which the molecule or structure passes.

iv Name the molecule which leaves the mitochondrion in order to provide energy for the cell.

10. One technique used to investigate the activity of cell organelles is called differential centrifugation. In this technique, a tissue is homogenized (ground in a blender), placed in tubes and spun in a centrifuge. This makes organelles sediment (settle) to the bottom of the tubes. The larger the organelles, the faster they sediment. By repeating the process at faster and faster speeds, the organelles can be separated from each other according to size. Some liver tissue was treated in this way to separate ribosomes, nuclei and mitochondria. The centrifuge was spun at 1000 g, 10 000 g or 100 000 g (g is gravitational force).

a State in which of the three sediments (1000 g, 10 000 g or 100 000 g) you would expect to find the following:

i ribosomes

ii nuclei

iii mitochondria

b Liver tissue contains many lysosomes. Suggest why this makes it difficult to study mitochondria using the differential centrifugation technique.

Answers:



TO BE MEMORIZED:

Membranous organelles:

ALL organelles are membranous, except:

- Ribosomes
- Centrioles
- Microtubules & Cytoskeleton

Organelles covered by double membrane:

- Mitochondria
- Nucleus
- Chloroplast

Structures containing DNA:

- Mitochondria (circular)
- Chloroplasts (circular)
- Prokaryotes & Viruses (circular)
- Nucleus (linear/chromatin)

Structures containing 70S ribosomes:

- Mitochondria
- Chloroplasts
- Prokaryotic cells

Can make proteins

Important diameter lengths:

- Cell membrane (7nm)
- Ribosomes (25nm)
- Mitochondria smaller than chloroplast
- Mitochondria same size as bacteria cells

Weird new names:

- Centrioles → produce spindle & MT, made of 9 triplets of MT
- Centrosome → MT organizing centre where centrioles produce MT
- Dynein → name of protein making inner and outer arms in cilia structure
- Actin → protein making microfilaments for cytoskeleton & microvilli
- Cisternae → description of the flattened sacs look of ER & Golgi body
- Cristae → description for folded inner membrane of mitochondria
- Histone → DNA coils around it to make chromatin → chromosomes
- Nucleolus → contains rRNA, site for production of ribosomes
- Lysosome → small vesicles containing hydrolytic enzymes, break down pathogens/old organelles/cell debris
- Plasmodesmata → cytoplasm connections between neighbouring plant cells, allows easy passage of substance/water
- 70S ribosomes → bacteria, mitochondria, chloroplasts
- 80S ribosomes→ plant, animal cell cytoplasm/rough ER

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Textbook summary:

SUMMARY

The basic unit of life is the cell. The simplest cells are prokaryotic cells, which are thought to have evolved before, and given rise to, the much more complex and much larger eukaryotic cells.
Cells can be seen clearly only with the aid of microscopes. The light microscope uses light as a source of radiation, whereas the electron microscope uses electrons. The electron microscope has greater resolution (allows more detail to be seen) than the light microscope because electrons have a shorter wavelength than light.
With a light microscope, cells may be measured using an eyepiece graticule and a stage micrometer. Using the formula $A = \frac{I}{M}$ the actual size of an object (A) or its magnification (M) can be found if its observed (image) size (I) is measured and A or M, as appropriate, is known.
All cells are surrounded by a partially permeable cell surface membrane that controls exchange between the cell and its environment. All cells contain genetic material in the form of DNA, and ribosomes for protein synthesis.
All eukaryotic cells possess a nucleus containing DNA. The DNA is linear (not circular) and bound to proteins and RNA to form chromatin.
The cytoplasm of eukaryotic cells contains many organelles, some of which are surrounded by one or two membranes. Organelles of eukaryotic cells include endoplasmic reticulum (ER), 80S ribosomes, Golgi apparatus, lysosomes and mitochondria. Animal cells also contain a centrosome and centrioles and may contain cilia. Plant cells have a cell wall containing cellulose. They may contain chloroplasts and often have a large central vacuole.
Prokaryotic cells lack a true nucleus and have smaller (70S) ribosomes than eukaryotic cells. They also lack membrane-bound organelles. Their DNA is circular and lies free in the cytoplasm.
Viruses do not have a cellular structure. They are extremely small and simple. They consist of a molecule of DNA or RNA, a protein coat and sometimes an outer envelope.